

Summary – House Price Prediction

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The correlation heat map shows that price has an approximate 50% correlation with area, bathrooms, number of stories, and air-conditioning status, indicating that these factors significantly influence house prices. In contrast, features such as bedrooms, parking, and other variables show weaker relationships with price.

The linear regression model achieved an R^2 of about 0.65, with a mean average error of around Rs.9.7 lakhs

The random forest model achieved an R^2 of about 0.61, with a mean average error of around Rs.10 lakhs

It is surprising to see random forest achieving lower performance compared to a simple model such as linear regression, the explanation to this lies in the fact that linear regression is a simple model and the dataset is considerable small, only 545 rows hence the more complex random forest model overfits on the training data and fails to perform on the test data a similar thing would have happened maybe if we employed a regression model of higher degree.

Another funny yet shocking thing is bathrooms correlated to price than a few other significant features which signifies that buyers value bathrooms than functional space? A point to be noted.

From a real estate business point of view prioritizing air conditioning, bathrooms and area count when selecting a property to market is crucial for a higher success rate with a greater resale value.